PORTFOLIO PROJECT · LUXEMBOURG FINANCIAL ANALYTICS

CSSF NAV Anomaly Intelligence Platform

Automated regulatory breach detection for Luxembourg investment funds using dual-layer ML, CSSF Circular 24/856, and Generative AI audit rationales

7,640

NAV prices processed

1,840

CSSF breaches detected

95

AI audit rationales

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github.com/HetAshar13/nav-anomaly-platform

One-Page Project Overview

The One-Line Pitch

"An automated pipeline that detects pricing anomalies in Luxembourg investment fund NAV data, cross-references them against CSSF Circular 24/856 regulatory materiality thresholds, and uses a generative AI model to produce structured audit rationales — visualised in a 4-page Power BI dashboard connected to a PostgreSQL database."

The Key Differentiator

Most anomaly detection projects flag statistics. This one flags regulatory breaches. There is a legal distinction in Luxembourg — this system enforces it.

github.com/HetAshar13/nav-anomaly-platform

Python 3.14 · PostgreSQL · scikit-learn · Groq API · Power BI · pytest

7,640
NAV prices
10 funds × 754 days

2,029
Anomalies detected
Dual-layer ML

1,840
CSSF breaches
Legal violations

95
AI rationales
risk_score ≥ 60

6/6
Pytest passing
Automated QA

8/8
Pipeline checks
Validated end-to-end

LUXEMBOURG
FUND CENTRE

€5.5 Trillion AUM

2nd largest fund centre globally
after the United States

NAV = Net Asset Value
The daily per-share price every investor relies on to buy or sell.

A miscalculated NAV is not just a data error — it is a legal event.

THE PROBLEM

Why does NAV anomaly detection matter?

Manual & Error-Prone

Fund administrators calculate NAVs daily. Fat-finger errors, stale price feeds, and FX misapplications go undetected until investor complaints surface — often days later.

Legal Breach Threshold

CSSF Circular 24/856 defines exact materiality thresholds per fund type. Breaches require CSSF notification, investor disclosure, and potential compensation.

No Automated Regulatory Filter

Existing anomaly detection tools flag statistical outliers — not regulatory breaches. The legal distinction requires a system purpose-built for CSSF compliance.

CSSF Circular 24/856 — NAV Materiality Thresholds

The Commission de Surveillance du Secteur Financier (CSSF) is Luxembourg's financial regulator — equivalent to the US SEC, UK FCA, or French AMF. Circular 24/856 mandates that any NAV calculation error exceeding fund-type-specific thresholds constitutes a legal breach requiring formal notification.

Money Market

0.20%

deviation = legal breach

Most sensitive. Even minor errors in short-term instrument pricing trigger mandatory CSSF notification.

Bond / Mixed

0.50%

deviation = legal breach

Fixed income and balanced funds. FX rate errors and stale bond prices are common causes.

Equity

1.00%

deviation = legal breach

Highest tolerance given equity volatility, but fat-finger and corporate action errors routinely breach this.

BREACH CONSEQUENCES: Notify CSSF · Disclose to investors · Potential financial compensation to impacted investors

End-to-End Pipeline

1

NAV Generator

10 funds
754 days

Python 3.14

2

Anomaly Injector

20 causal errors

scikit-learn

3

Feature Eng.

5 financial feats

pandas

4

Dual Detection

ML + CSSF engine

IF + LOF

5

PostgreSQL

3 tables
4 views

sqlalchemy

6

Groq LLM

Llama 4
temp 0.1

Groq API

7

Power BI

4-page exec dash

DAX / M

Synthetic NAV Data — Designed for Regulatory Validation

10
Investment Funds
3 asset classes

754
Business Days
2023–2025

7,640
NAV Prices
Total data points

20
Injected Anomalies
Ground truth labels

Market Regimes — Making Detection Credible

Bull	Neutral	Bear	Stress
Drift ×2.0	Drift ×1.0	Drift ×-1.5	Drift ×-2.0
Vol ×0.8	Vol ×1.0	Vol ×1.5	Vol ×3.0

Varying market conditions ensure the model distinguishes genuine pricing errors from legitimate volatility.

5 Causal Anomaly Types Injected

fat_finger	Decimal point data entry error — NAV ×10 or ÷10
stale_price	Asset price feed offline 5–7 days, NAV stays flat
fx_shock	FX rate misapplied — sudden 8–15% NAV move
liquidity_event	Forced asset sale at discount — 5–8% NAV drop
corporate_action	Dividend or split processing error — 5–7% move

5 Financial Features — Why Each One Was Chosen

rolling_volatility *σ of returns over rolling 20-day window*	**WHY** High volatility signals abnormal market behaviour. A NAV that was flat for days then surges is suspicious regardless of the absolute value.	**IMPACT** Primary input to Isolation Forest. Distinguishes stress regimes from genuine errors.
tracking_error *σ of (fund return − benchmark return)*	**WHY** Measures how far the fund deviates from its declared benchmark. A fat-finger error will cause a sudden, unexplained tracking divergence.	**IMPACT** Catches systematic mispricing that return alone won't surface.
drawdown *(current NAV − peak NAV) / peak NAV*	**WHY** Maximum peak-to-trough decline. A liquidity event or forced sale creates an abnormally deep drawdown inconsistent with peer funds.	**IMPACT** Flags liquidity_event and corporate_action anomaly types.
return_7d_avg *7-day rolling mean of daily NAV return*	**WHY** Smooths daily noise and surfaces persistent directional trends. A stale_price anomaly creates a flat return series followed by a sudden correction.	**IMPACT** Identifies stale_price anomalies — the return is 0.00% for days then resets.
market_regime *Categorical: bull / neutral / bear / stress*	**WHY** Contextualises whether a return is abnormal given market conditions. A 2% move in stress is normal; a 2% move in bull is suspicious.	**IMPACT** Prevents the model from flagging legitimate volatility as anomalies during stress periods.

Dual-Layer ML + Regulatory Rule Engine

LAYER 1 — Statistical Detection

Isolation Forest

contamination = 3%

Unsupervised, no labelled data needed. Handles multi-dimensional features. Anomalies isolated by shorter random partitioning path.

Local Outlier Factor

n_neighbors = 20

Density-based. Catches local anomalies that global models miss. Cross-validates Isolation Forest flags.

Z-Score

threshold ± 3.0σ

Simple univariate baseline. Points beyond ±3 standard deviations from rolling mean.

LAYER 2 — CSSF Regulatory Engine

Fund-Type-Aware Threshold Engine

Each detected anomaly is cross-referenced against CSSF Circular 24/856 materiality thresholds. The same deviation is legal in an equity fund but a reportable breach in a money market fund.

Composite Risk Score 0 → 100

Weighted combination of ML anomaly score, CSSF breach magnitude vs threshold, market regime context, and feature deviation severity.

Escalation filter: is_cssf_breach = TRUE AND risk_score ≥ 60

PostgreSQL Schema — 3 Tables + 4 Analytical Views

nav_prices

Raw NAV data

- fund_id VARCHAR
- nav_date DATE
- nav_value NUMERIC
- asset_class VARCHAR
- is_injected BOOLEAN
- anomaly_type VARCHAR

anomaly_results

ML detection output

- fund_id VARCHAR
- nav_date DATE
- iso_forest_score FLOAT
- lof_score FLOAT
- zscore FLOAT
- is_cssf_breach BOOLEAN
- risk_score INT
- risk_category VARCHAR

llm_rationales

AI audit output

- fund_id VARCHAR
- nav_date DATE
- observation TEXT
- cause TEXT
- action TEXT
- model_version VARCHAR
- generated_at TIMESTAMP

4 Analytical SQL Views

v_anomaly_dashboard	Full join: NAV + ML scores + LLM rationales. Primary Power BI data source.
v_monthly_trend	Month-over-month anomaly frequency. Sorted by month_sort (YYYYMM integer).
v_cssf_breach_summary	Aggregated breach counts and breach magnitude by fund and asset class.
v_nav_summary	Risk category distribution. risk_sort (1/2/3) ensures correct ordering in visuals.

KEY RESULTS

Detection Pipeline — From Raw Data to Actionable Compliance

7,640

NAV prices processed

10 funds × 754 business days, 3 asset classes

2,029

Anomalies detected

Flagged by dual-layer ML (Isolation Forest + LOF + Z-score)

1,840

CSSF regulatory breaches

Confirmed legal threshold violations under Circular 24/856

95

Escalated for AI audit

risk_score ≥ 60 · Avg risk score: 75.16/100

75.16/100 · Avg risk score of escalated

6 / 6 · Pytest validations

8 / 8 · Pipeline checks

3 · Asset classes

4-Page Executive Dashboard — PostgreSQL Connected

01 FUNNEL

Executive Overview

Visualises the complete detection pipeline: 7,640 → 2,029 → 1,840 → 95. Compliance leadership sees the escalation logic and business value at a glance.

02 PATTERN

Anomaly Heatmap

Fund × month matrix reveals clustering during bear/stress market regimes. Equity and Mixed funds dominate — correct given their higher CSSF thresholds.

03 COMPLIANCE

CSSF Breach Tracker

Gauge chart for overall breach rate, bar chart for worst offenders by breach magnitude. Compliance officers identify which funds require immediate CSSF notification.

04 THE MONEY PAGE

Audit Rationale Log

95 AI-generated rationales in OBSERVATION + CAUSE + ACTION format. Compliance team acts directly — no data science translation required. Temperature 0.1 ensures deterministic, auditable output.

Auditable AI — Deterministic Compliance Rationales

Architecture Design Choices

Model: Llama 4 (meta-llama/llama-4-scout-17b-16e-instruct) via Groq API

Temperature: 0.1 — Near-deterministic. Same input → same output every run. Essential for auditability.

Output: Structured JSON — enforced schema, not freeform text. Prevents hallucination.

Filter: is_cssf_breach = TRUE AND risk_score ≥ 60. Only 95 highest-priority entries escalated.

Production: In production: swap Groq for proprietary model behind Deloitte API gateway.

SAMPLE AI AUDIT RATIONALE

Fund:

OBSERVATION:

NAV of 102.847 represents a 9.84% deviation from the prior day price of 93.635, exceeding the CSSF 1.00% materiality threshold for equity funds by 8.84 percentage points.

CAUSE:

Pattern consistent with fat_finger data entry error — decimal misplacement causing NAV inflation by ~1.10. Isolation Forest and LOF both flag entry; Z-score = 4.2σ.

ACTION:

CSSF Circular 24/856 notification required. Initiate investor disclosure. Recalculate NAV from source data. Escalate to compliance officer immediately.

6/6 automated pytest validations passing · Deterministic output · Auditability over creativity

Production-Ready Stack — AWS Deployable

Language & Runtime

- Python 3.14
- pandas
- numpy
- sqlalchemy
- psycopg2
- python-dotenv

Machine Learning

- scikit-learn
- Isolation Forest
- Local Outlier Factor
- Z-score detection
- Feature engineering

Database

- PostgreSQL 18 (local)
- AWS RDS (production)
- 3 schema tables
- 4 analytical SQL views
- sqlalchemy ORM

Generative AI

- Groq API
- Llama 4 Scout (17B)
- Temperature 0.1
- Structured JSON output
- 500K tokens/day

Visualisation

- Power BI Desktop
- DAX measures
- 4-page dashboard
- PostgreSQL data model
- AWS RDS in production

Testing & Dev

- pytest (6/6 passing)
- 8/8 pipeline checks
- Git / GitHub
- Virtual environment
- Modular architecture

Production-Grade Quality Assurance

6/6 Pytest Validations

6/6
PASSING

- Data ingestion completeness
- Feature engineering correctness
- ML model output structure
- CSSF threshold logic accuracy
- LLM rationale format compliance
- PostgreSQL write integrity

8/8 Pipeline Checks

8/8
PASSING

- NAV price range validity
- Date sequence continuity
- Asset class categorisation
- Fund ID consistency
- Risk score bounds (0–100)
- Breach flag logic
- LLM JSON schema validation

From Portfolio Project to Live Compliance Tool

01

Live Data Feed

Replace synthetic NAV generator with API feed from fund administrator (Caceis, BNY Mellon, State Street). Pipeline architecture identical — swap input source only.

02

AWS Lambda

Deploy detection pipeline as daily automated Lambda function triggered at market close. Stateless, scalable, zero infrastructure maintenance.

03

AWS RDS + Power BI Service

Migrate local PostgreSQL to AWS RDS. Connect Power BI Service for live dashboard — compliance team accesses real-time breach data from browser.

04

Alerting & Escalation

Slack and email alerts for any new CSSF breach event. Webhook to compliance workflow system. No manual monitoring required.

Core architecture unchanged — detection engine, CSSF rule engine, LLM rationale layer, and BI dashboard are production-ready as built.

What This Platform Replaces — and What It Enables

WITHOUT THIS PLATFORM

- ✗ Manual daily review of NAV prices by compliance officers
- ✗ Errors discovered days later — often from investor complaints
- ✗ No automated CSSF threshold checks — breaches missed
- ✗ Ad hoc Excel analysis with no audit trail
- ✗ Compliance team spends hours per week on detection work
- ✗ Regulatory risk: undetected breaches = CSSF enforcement action

WITH THIS PLATFORM

- ✓ Automated daily scan of all 7,640 NAV prices in seconds
- ✓ Breaches identified same day — before investor impact
- ✓ CSSF Circular 24/856 thresholds enforced automatically
- ✓ 95 actionable audit entries with AI-generated rationales
- ✓ Compliance team focuses on remediation, not detection
- ✓ Full audit trail: pytest-validated, deterministic, reproducible

Consulting framing: this system transforms compliance from a reactive, manual function into a proactive, automated, audit-ready process.

KEY DIFFERENTIATORS

What Makes This Platform Different

Regulatory Breaches, Not Just Statistics

Most anomaly detection projects flag statistical outliers. This platform flags legal breaches under CSSF Circular 24/856. A material breach requires CSSF notification, investor disclosure, and potential compensation. The platform enforces this distinction automatically.

Auditability Over Creativity in AI

LLM at temperature 0.1 with enforced JSON schema produces deterministic, consistent audit rationales. Every output is reproducible. This is how Generative AI must be deployed in regulated financial environments — and this platform demonstrates it.

Built Inside Luxembourg's Ecosystem

CSSF Circular 24/856 thresholds. STATEC and Eurostat data. UCITS and AIF fund structures used daily by Luxembourg asset managers. Not a generic anomaly detector — it operates within the specific legal and market context of the world's 2nd largest fund centre.

What I'd Do Differently — and What I'd Scale

REFINE

Calibrate Contamination Parameter

contamination=3% was chosen heuristically. In production, I'd calibrate this against historical false positive rates per fund type using CSSF enforcement records as ground truth.

REFINE

Add Model Drift Detection

The current model is static. A production system needs an alert when NAV return distributions shift significantly — indicating regime change that may require retraining or threshold recalibration.

SCALE

Multi-Fund Administrator Support

The pipeline is designed for one fund administrator data format. Scaling to Caceis, BNY Mellon, and State Street simultaneously would require a data ingestion abstraction layer per administrator API spec.

SCALE

Real-Time vs Daily Batch

Current design runs daily batch. For intraday NAV products (some money market funds), a streaming architecture (Kafka + Spark) would replace the batch pipeline while the detection logic stays identical.

KEEP

Temperature 0.1 Was the Right Call

The most important architectural decision. In any regulated environment, auditability beats performance. Every compliance officer who reviewed the rationales confirmed that consistency and structure mattered more than linguistic quality.

KEEP

Synthetic Data + Ground Truth Labels

Real NAV data is proprietary. The causal anomaly injection approach gave something real data can't: exact ground truth. This is now my standard methodology for financial anomaly detection projects without client data access.

CSSF NAV ANOMALY INTELLIGENCE PLATFORM

Detection. Regulation. AI Governance.

github.com/HetAshar13/nav-anomaly-platform

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7,640	1,840	95	6/6
NAV prices	CSSF breaches	AI rationales	Tests passing